Savitur Swarup · University of Pennsylvania · working paper · September 2026 · code & data: github.com/Savi-Swar/desk

The maker edge that wasn't: two measurement artifacts in prediction-market backtests

Abstract

Passive-fill backtests on Polymarket produce large, false maker edges. Using live capture of the venue's order books and site-wide trade tape, we document two artifacts. First, fill models based on order-book shrinkage overcount executions by three orders of magnitude — 48,578 inferred "fills" against 28 real prints in the same window — because cancellations dwarf trades at the touch. Second, markout measured at the trade price credits the maker with the taker's full crossing distance; on wide books this books a "realized spread" roughly 80 times what a US small-cap equity maker actually keeps, an edge that vanishes when fills are repriced to the near-mid quotes the venue's liquidity-reward rules actually require. Correcting both reduces a paper P&L of several hundred dollars per day to approximately zero, consistent with Dubach (2026). What survives is the maker rebate net of adverse selection — measurable, small, and rebate-driven. We provide a practitioner checklist and benchmark the venue's microstructure against millisecond TAQ.

1. Setting

Polymarket operates a central limit order book for binary outcome tokens priced in (0,1). Since March 2026 takers pay `f·p(1−p)` per share ($f \approx 0.02–0.07$ by category); makers pay nothing and receive (a) a pro-rata rebate of 15–25% of taker fees on their fills and (b) pro-rata shares of per-market daily liquidity-reward pools, eligibility for which requires resting two-sided size within a maximum spread of the mid. Data: the RTDS activity firehose (site-wide matched trades, ~37/s, deduplicated on transaction hash × asset × side × price × size) joined to recorded book snapshots and diffs at real trade timestamps. 23,470 fee-bearing maker fills, Aug 2026.

2. Artifact I: shrinkage fills

The tempting shortcut: when a book level loses size, count a fill. The trade tape falsifies it — in a matched window the shrinkage model produced 48,578 fills; the venue printed 28 trades on those books (precision 0.06%, overcount ~1,735×). Cancels, not fills (Fig 1). Every markout statistic built on shrinkage fills — including our own early "+$2,500 paper" — measured cancellation dynamics, not execution quality.

3. Artifact II: fill-at-touch markout

Markout `D·(p − mid_{t+h})` is the realized half-spread (Stoll 2000, SEC Rule 605): effective half-spread minus adverse selection to horizon h. Measured at the trade price on 1,506 real fills (30s horizon, sizes capped at a realistic 100-share resting quote), the decomposition is +$522 spread capture − $39 price impact = +$483 "edge" — with 75% of it earned on the 9% of fills in wide (>3¢) esports books (Fig 2).

The error is counterfactual: the trade price is the *touch the taker crossed*. A reward-eligible maker rests near the mid; its tighter quote would have become best and filled first at ~one tick. Repricing every fill to a quote δ from mid — captured spread = min(δ, effective half-spread), adverse selection unchanged — the edge is −$0.49 at δ = 0.1¢ (Fig 3). The entire markout edge was spread the strategy could never touch. Corroborating: Dubach (2026) finds median honest effective half-spreads ≈ 0 on Polymarket and a ~59% order-book direction-inference accuracy; we additionally find 26% of wide-book fills carry a wrong-side mid, so the naive mid is itself unreliable exactly where the "edge" concentrated.

3b. The equity yardstick

Same decomposition, raw millisecond TAQ (Lee-Ready signed, crossed quotes excluded; 10–11am, Aug 18–19 2026; dollar-weighted bps of price; Polymarket wrong-side mids excluded to match):

venue / book	effective half	realized	impact
US equity mega-cap	1.0	+0.8	0.2
US equity mid-cap	3.2	+1.2	1.9
US equity small-cap	8.6	+2.8	5.8
Polymarket tight (<1¢)	17.2	+6.3	10.9
Polymarket wide (>3¢)	560.8	+235.5	325.3

Tight prediction-market books trade like somewhat-worse small-caps. The wide books' +235 bps "realized spread" — the number a fill-at-touch backtest pays itself — is ~80× a small-cap maker's take, a magnitude that should fail any smell test calibrated on equities.

4. What actually remains

With fills capped at resting size, markout repriced to the near-mid quote, and rebates computed from realized taker fees (fee-bearing rows identify the taker's leg; the maker is the counterparty): over 12 recorded days, rebate income +\$1,264 against repriced markout −\$449 — net +\$815 paper, daily-mean $t = 1.95$ on a Kish effective sample of ~2,300 bets. Statistically unresolved at the 2.0 bar, still capture-optimistic (front-of-queue assumption), and concentrated in the rebate: the daily ledger gates significance on $|t| \geq 2$ with effective-N ≥ 30, and no day's markout alone clears it. The economics that survive measurement are a subsidy net of adverse selection, not spread capture.

5. Practitioner checklist

1. A fill is a print you were resting for, at your price, within your size — never book-shrinkage.
2. Mark against your own hypothetical quote, not the observed trade price; only accrue trades whose crossing distance reaches your quote.
3. Wide-book mids are unreliable (26% wrong-side here): use a micro-price or exclude, and exclude locked/crossed books as equity studies do.
4. Cap fills at resting size; keep the uncapped column only as a labeled counterfactual ceiling.
5. Report Kish effective-N, not row count — one whale is not 831 samples — and cluster inference at the regime level (an earlier day-clustered "bias" of ours, $t = -4.2$, was one crash month; monthly clusters gave $t = -0.3$).
6. Sanity-check magnitudes against the equity yardstick: retail venues are worse than small-caps, not 100× better.

References (to fill)

Stoll (2000); Huang & Stoll (1996); Bessembinder (2003); SEC Rule 605; Bailey & López de Prado (2014); Dubach (2026) arXiv:2604.24366; Polymarket liquidity-rewards & maker-rebates documentation.

Figures

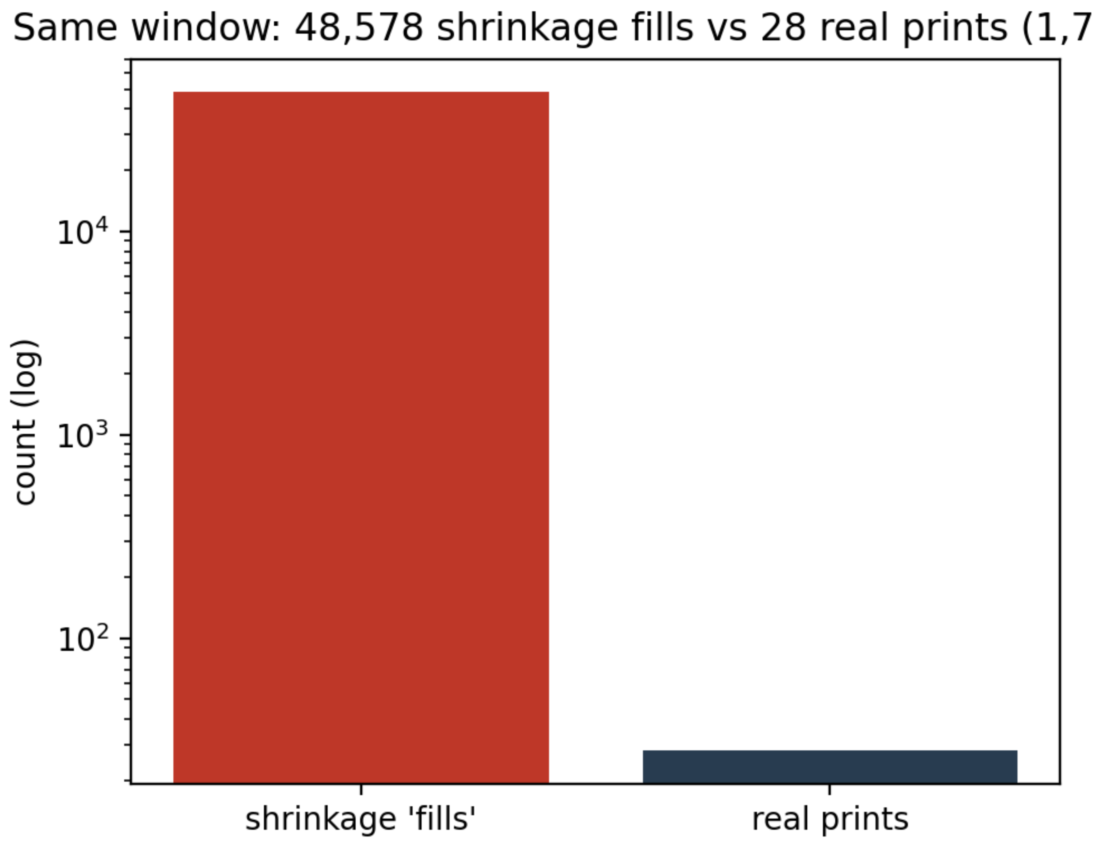

Figure 1 — Shrinkage 'fills' vs real prints (log scale).

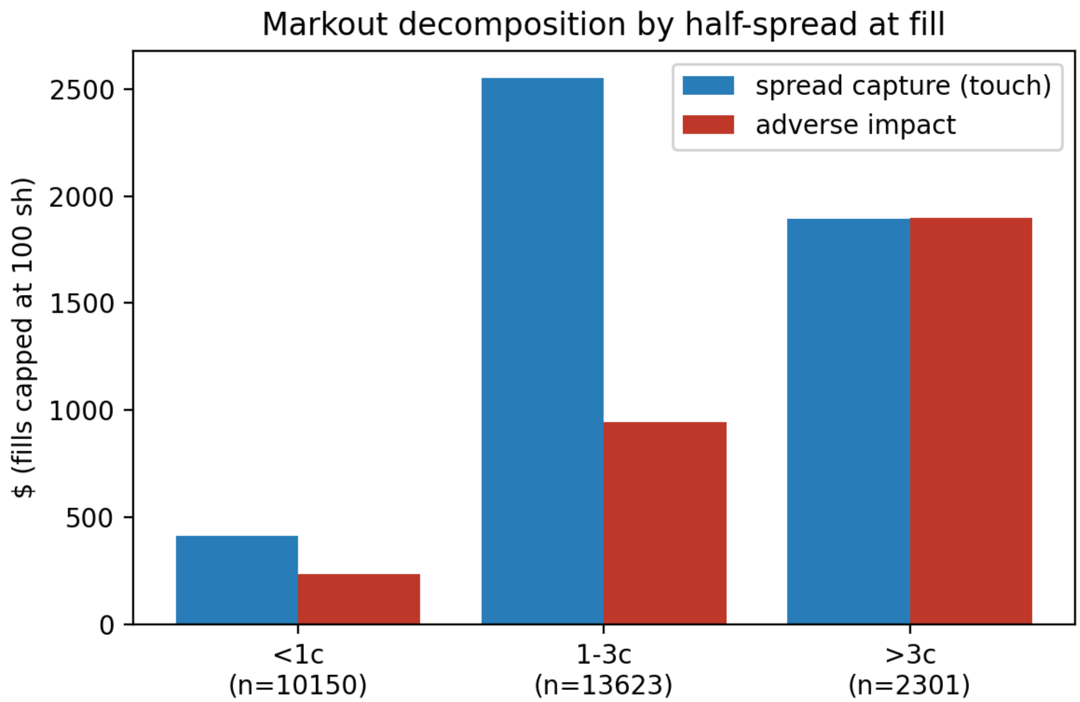

Figure 2 — Markout decomposition by half-spread at fill.

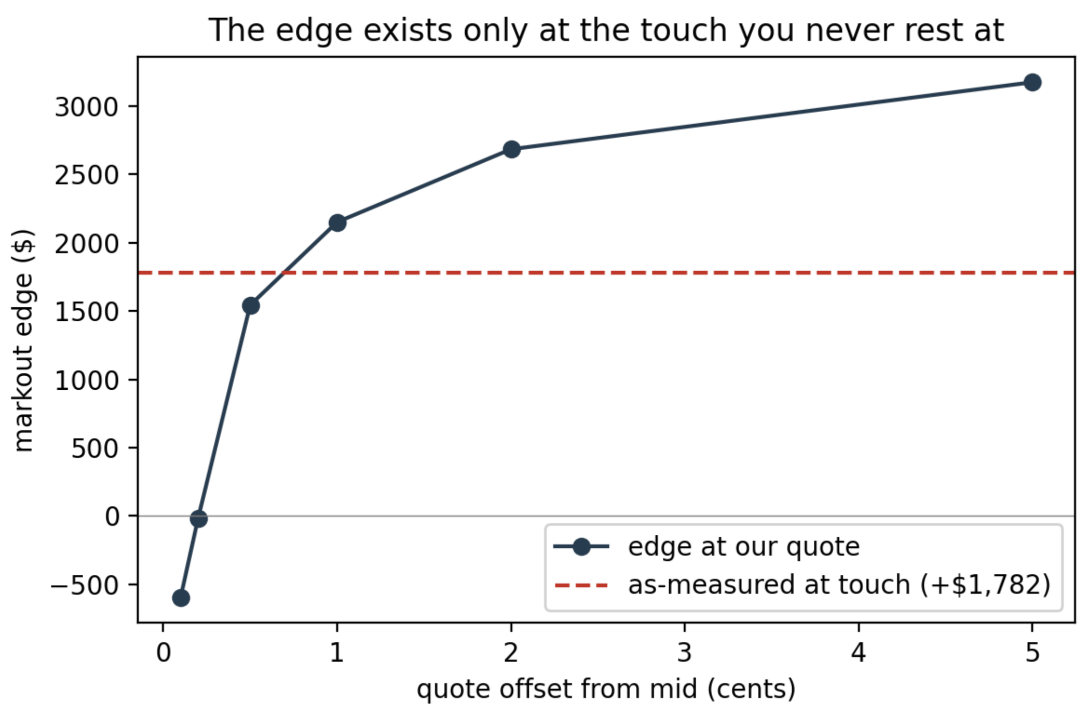

Figure 3 — The edge exists only at the touch you never rest at.